

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/01

Paper 1 Multiple Choice

25 September 2013

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

This document consists of **27** printed pages and **1** blank page.



1 Which molecules, found in cell surface membranes, contribute to cell recognition?

- A** Cholesterol and glycolipids
- B** Cholesterol and phospholipids
- C** Glycolipids and glycoproteins
- D** Phospholipids and glycoproteins

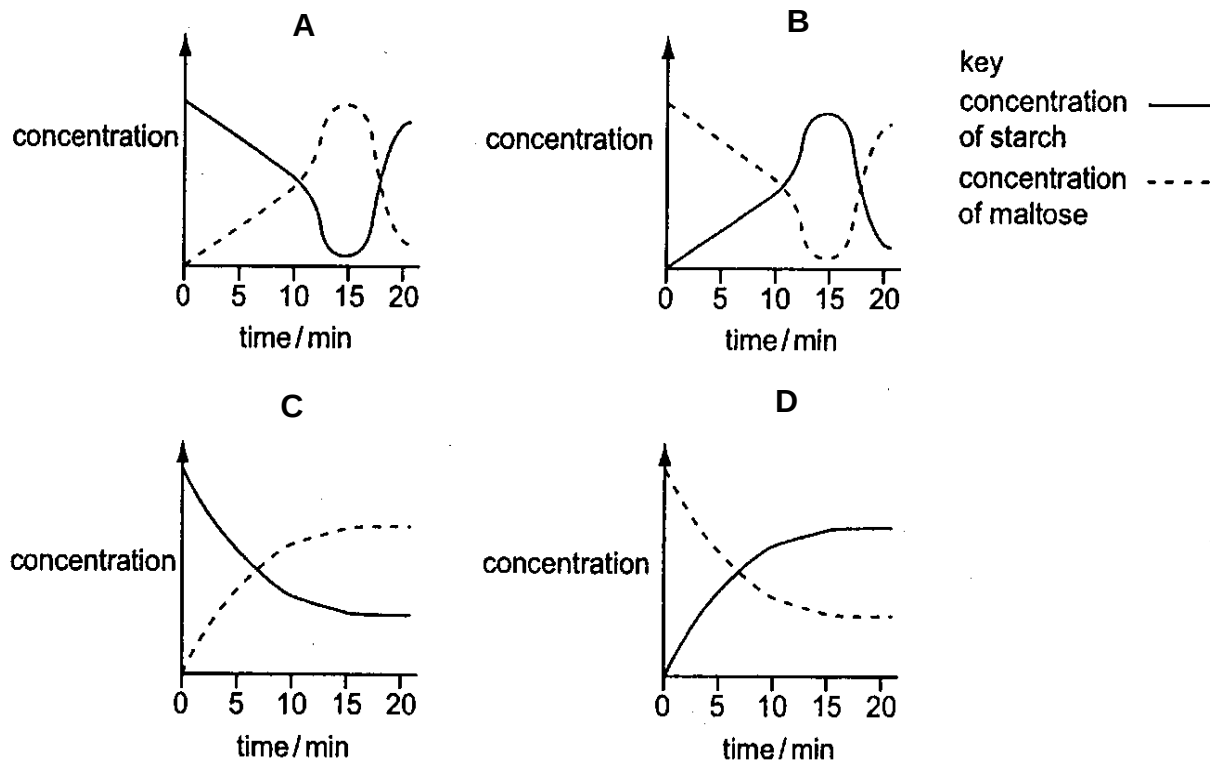
2 An enzyme is a globular protein held together by several different types of bonds, giving the enzyme primary, secondary and tertiary levels of structure.

Which combination correctly summarizes the types of bond involved in each level of structure?

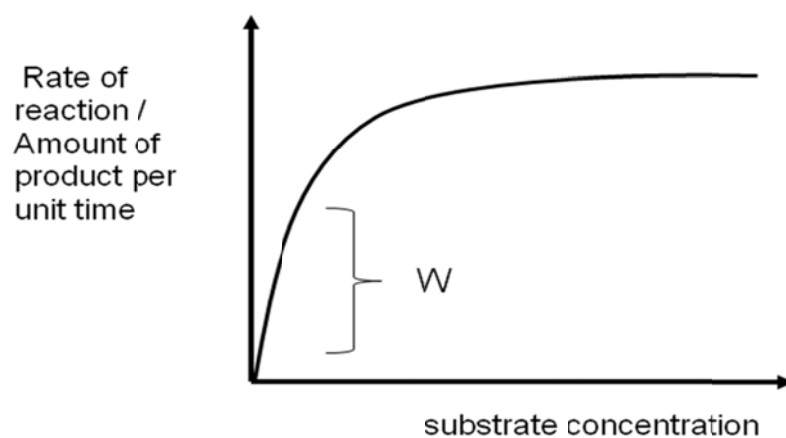
	Disulfide bonds	Hydrogen bonds	Ionic bonds	Peptide bonds
A	Primary	Tertiary	Secondary	Tertiary
B	Primary, tertiary	Primary	Secondary, tertiary	Secondary
C	Secondary	Secondary	Tertiary	Primary
D	Tertiary	Secondary, tertiary	Tertiary	Primary

- 3 In an experiment using starch and amylase, the concentrations of starch and maltose present in the reacting mixture are measured every minute for 20 minutes. 1% hydrochloric acid is added after 10 minutes and the mixture is heated to 60°C at 14 minutes.

Which graph represents the results of this experiment?



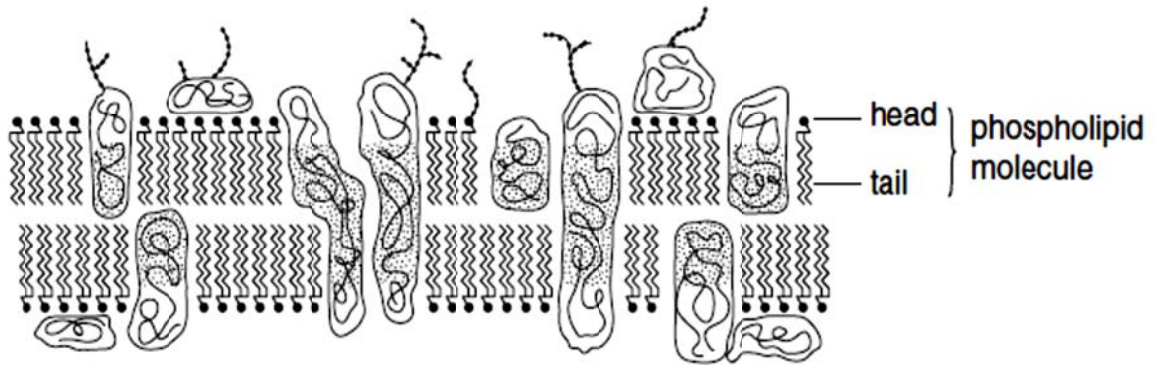
- 4 The graph below shows the rate of an enzyme catalyzed reaction with increasing substrate concentration. The reaction is carried out at 37°C and a pH of 7 for all substrate concentrations.



Which of the following(s) would result in a decrease in the rate of reaction at W?

- 1 Decrease in enzyme concentration
 - 2 Decrease in temperature to 27°C
 - 3 Increase in pH to 9
 - 4 Addition of competitive inhibitor
- A** 1 and 4 only
B 2 and 3 only
C 2, 3 and 4 only
D 1, 2, 3 and 4

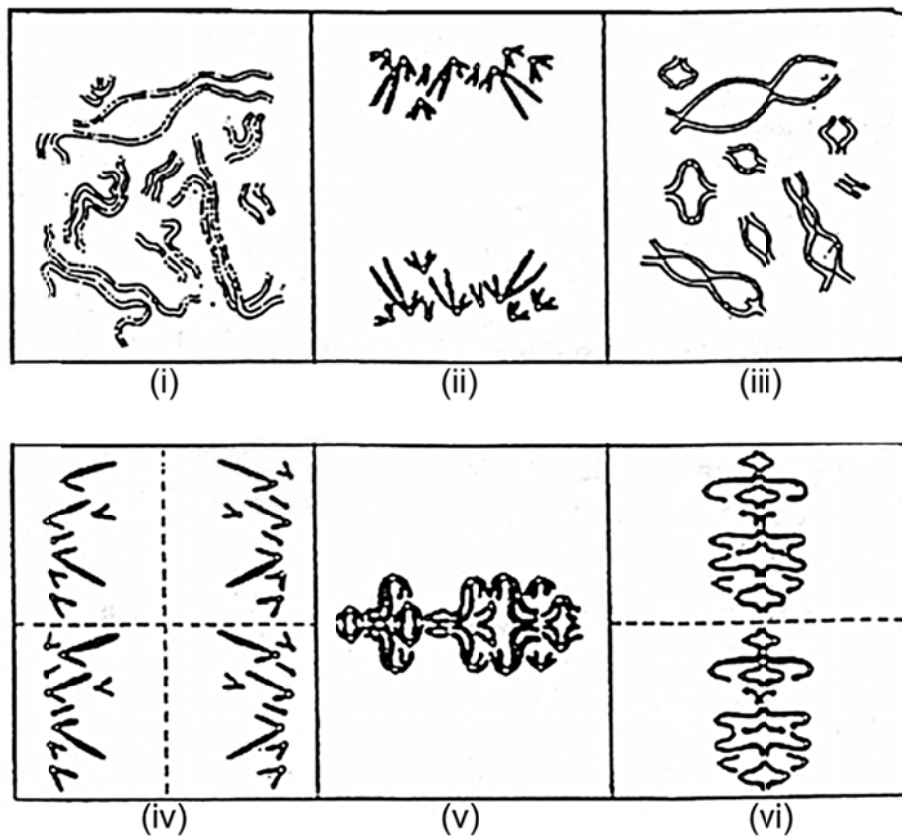
- 5 The diagram shows a section of a cell surface membrane.



What causes the phospholipid molecules to be arranged as shown?

- A The heads are hydrophilic and point towards water molecules.
 - B The heads are hydrophobic and point away from water molecules.
 - C The tails are hydrophilic and point away from water molecules.
 - D The tails are hydrophobic and point towards water molecules.
- 6 Which pair(s) of factors is inversely proportional to the rate of diffusion?
- 1 Concentration gradient and surface area over which diffusion occurs.
 - 2 Distance over which diffusion occurs and size of diffusing molecule.
 - 3 Size of diffusing molecule and concentration gradient.
 - 4 Surface area over which diffusion occurs and distance over which diffusion occurs.
- A 2 only
 - B 1, 2 and 4 only
 - C 2, 3 and 4 only
 - D All of the above

- 7 The figure below shows 6 stages of meiosis occurring in a plant cell, where $2n = 18$.



What is the correct order of these 6 stages?

- A (iii), (v), (ii), (vi), (iv), (i)
 B (iii), (i), (v), (ii), (vi), (iv)
 C (ii), (iii), (i), (v), (vi), (iv)
 D (i), (iii), (v), (ii), (vi), (iv)
- 8 A cell containing 92 chromatids at metaphase of mitosis would produce two nuclei each containing ____ chromosomes?
- A 92
 B 46
 C 23
 D 16

- 9** Cancer is a multi-step process. Some of the stages in cancer are listed.
- 1 Mutated cell undergo uncontrolled cell division by mitosis.
 - 2 Formation of tumour.
 - 3 Mutations in tumour suppressor genes and proto-oncogenes.
 - 4 Cells from the primary tumour metastasize to other sites via the bloodstream.

What is the correct sequence of events?

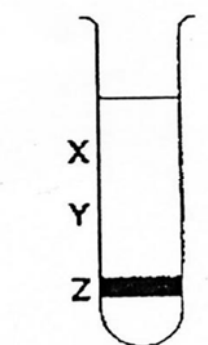
- A** 1 → 4 → 3 → 2
- B** 1 → 2 → 3 → 4
- C** 3 → 1 → 2 → 4
- D** 3 → 2 → 1 → 4
- 10** Cells taken from a human bone cancer multiplied readily in culture. Analysis showed that the cells were unable to produce the protein, RB. Addition of RB to these cells reduced their rate of division.

Which of the following statement accounts for the observation?

- A** Both chromosomes in the cancer cell carry alleles for tumour suppressor gene.
- B** Both chromosomes in the cancer cell have the allele for tumour suppressor gene deleted.
- C** Both chromosomes in the cancer cell carry alleles for proto-oncogene.
- D** Both chromosomes in the cancer cell have the allele for proto-oncogene deleted.

- 11** A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N . The culture was then allowed to reproduce using nucleotides containing normal ^{14}N . The DNA was examined using a centrifuge after one generation and again after two generations.

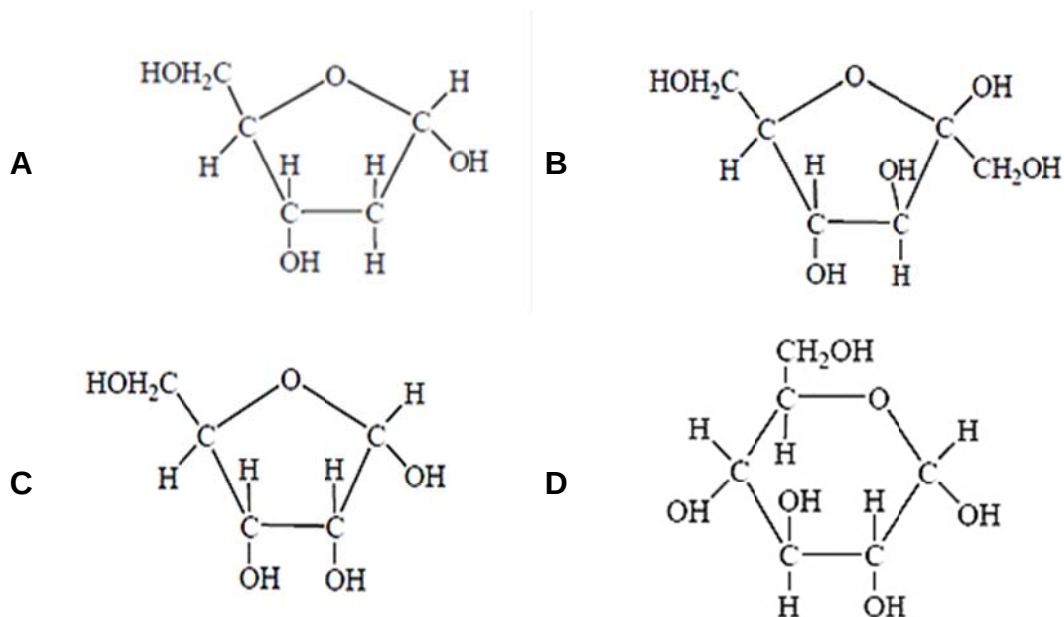
The diagram shows the position of the DNA band at Z in the centrifuge tube when the DNA was first labelled.



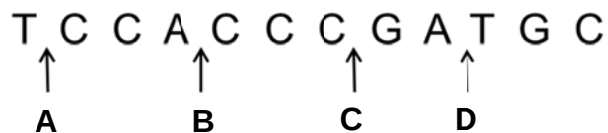
In which pattern would the DNA be found after the first and after the second cell generations?

	After first generation	After second generation
A	Half at X and half at Y	Quarter at X and at Z and half at Y
B	Half at X and half at Z	Quarter at X and at Z and half at Y
C	All at X	Half at X and half at Y
D	All at Y	Half at X and half at Y

- 12** Which of the following diagrams is the structure of a ribose sugar?



- 13** Which of the following statement is true of the end-replication problem of linear DNA?
- 1 When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
 - 2 Telomerase prevents the end-replication problem from occurring.
 - 3 Repeated rounds of replication produce shorter and shorter DNA molecules.
 - 4 Prokaryotes do not have the end-replication problem.
- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4
- 14** If there were 35 amino acids and DNA only contained three types of nitrogenous bases, what would be the minimum number of bases per codon that could code for proteins?
- A** 3
B 4
C 6
D 8
- 15** The RNA triplet UAG acts as a stop codon which terminates the synthesis of a polypeptide. The diagram shows a strand of DNA which codes for four amino acids. In which position will an insertion of a thymine nucleotide result in the termination of translation?



- 16** Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

A genetic code, showing mRNA codons, is shown below.

first position	second position				third position
	U	C	A	G	
U	phe	ser	tyr	cys	U
	phe	ser	tyr	cys	C
	leu	ser	STOP	STOP	A
	leu	ser	STOP	trp	G
C	leu	pro	his	arg	U
	leu	pro	his	arg	C
	leu	pro	gln	arg	A
	leu	pro	gln	arg	G
A	ile	thr	asn	ser	U
	ile	thr	asn	ser	C
	ile	thr	lys	arg	A
	met	thr	lys	arg	G
G	val	ala	asp	gly	U
	val	ala	asp	gly	C
	val	ala	glu	gly	A
	val	ala	glu	gly	G

Which event occurs to explain the information given above?

- A** There is no tRNA with an anticodon complementary to the first codon.
- B** The first codon is removed from the mRNA transcript in post-transcriptional modification.
- C** The mRNA binds to the rRNA in the second codon position.
- D** The first amino acid on the polypeptide chain is removed in post-translational modification.

17 Which of the following statements are correct?

- 1 A mutation in an allele occupying a gene locus may result in a change in the individual's phenotype.
- 2 Gene mutations always give rise to recessive alleles.
- 3 Chromosomal mutations can give rise to new alleles.
- 4 Aneuploids have extra chromosomes in their genome.

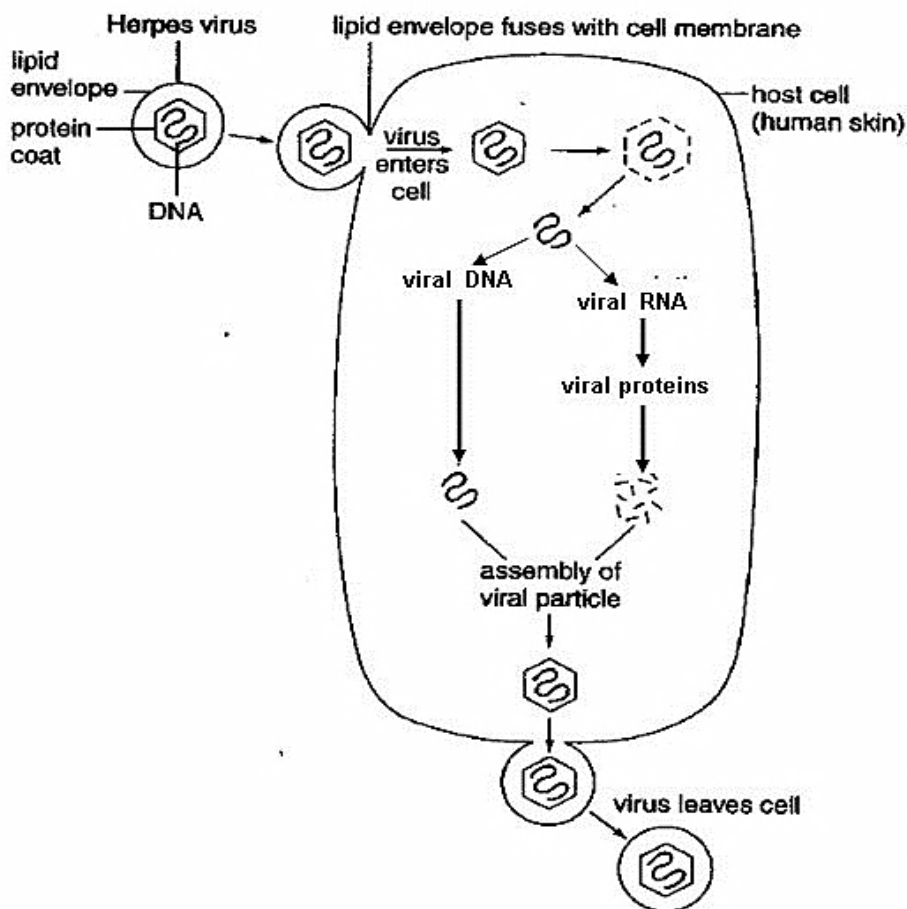
A 1 only

B 1 and 4

C 1, 3 and 4

D 2, 3 and 4

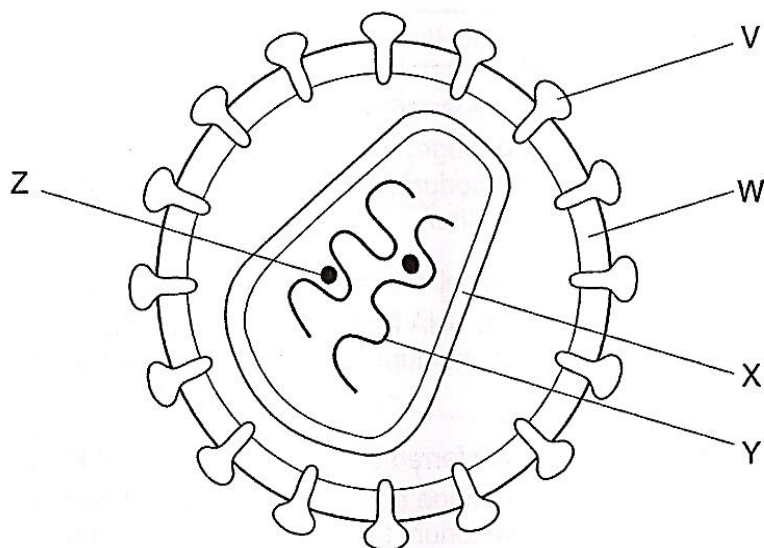
- 18 The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth.



With reference to the diagram above, which of the following statements **best** describes the herpes virus?

- A It is not a retrovirus as it contains DNA as its genetic material.
- B Its mode of replication is similar to that of HIV.
- C Its replication cycle includes a lysogenic phase.
- D Death of the host cell is necessary for the release of the viral progeny.

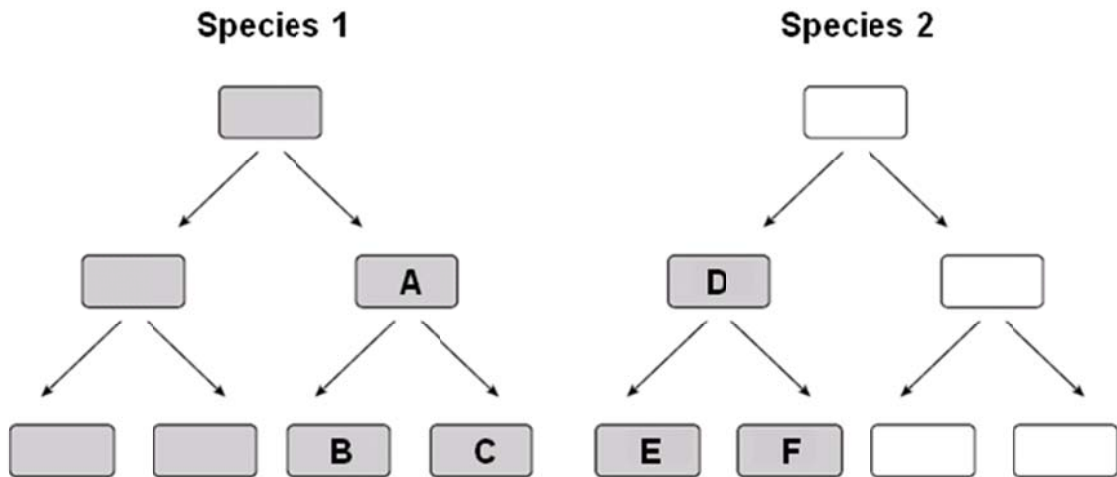
- 19 The diagram represents the structure of the human immunodeficiency virus (HIV).



Which combination correctly identifies the chemical constituents of the lettered components?

	V	W	X	Y	Z
A	amino acids	amino acids and carbohydrates	lipids	nucleotides	amino acids
B	amino acids	lipids	amino acids	amino acids	amino acids and carbohydrates
C	amino acids and carbohydrates	lipids	amino acids	nucleotides	amino acids
D	amino acids and carbohydrates	amino acids	lipids	nucleotides	amino acids and carbohydrates

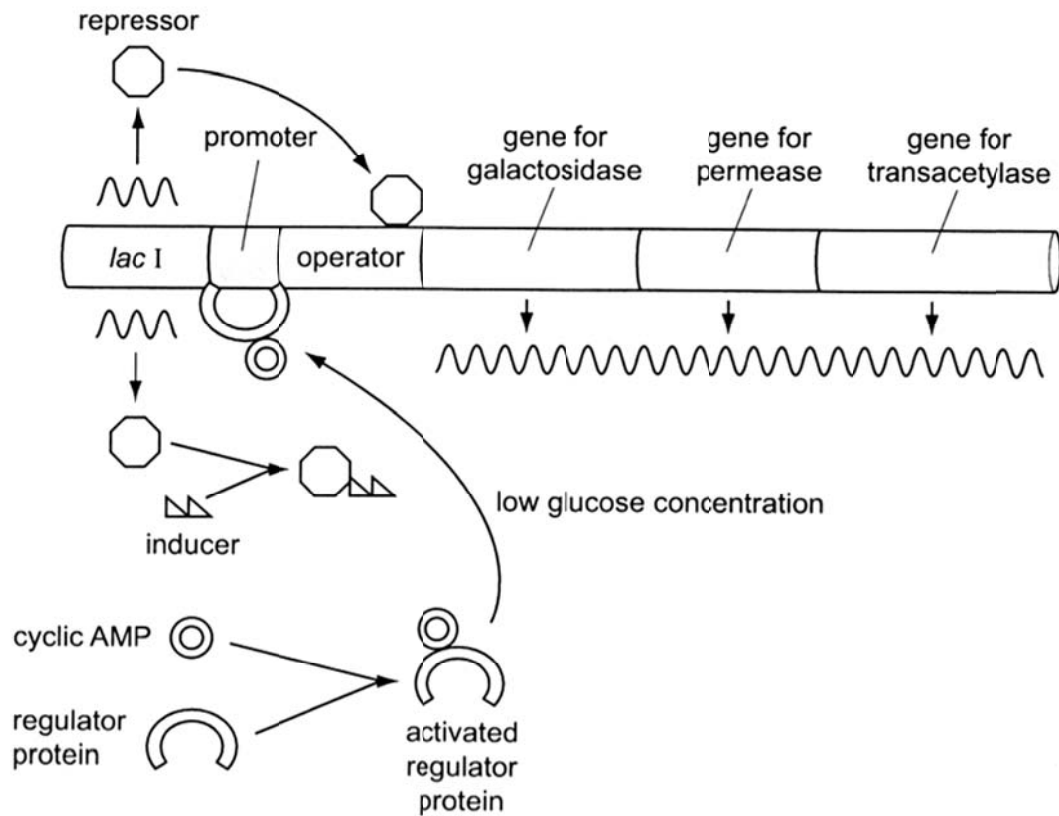
- 20 The diagram below shows 2 species of bacteria. The bacteria that are shaded are resistant to the antibiotic penicillin.



Which one of the following statements is likely to be true?

- A** Bacterium D is resistant to penicillin due the transfer of the complete F plasmid and penicillin resistant gene through transduction involving Bacterium A.
- B** Bacterium D is resistant to penicillin due to horizontal gene transfer involving a bacteriophage and Bacterium A.
- C** Bacterium E acquired resistance to penicillin as a result of random mutation and genetic transformation.
- D** Bacteria E and F are resistant to penicillin as a result of conjugation from Bacterium D.

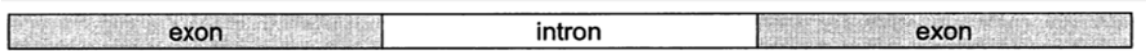
21 The diagram shows some of the factors affecting the lac operon.



Which statement describes the role of the regulator protein in the expression of the lac operon?

- A Activated regulator binding stabilizes the transcriptional complex.
- B Activated regulator binding to the operator increases mRNA formation.
- C Inactive regulator inhibits transcription.
- D Regulator protein is activated when ATP level is high.

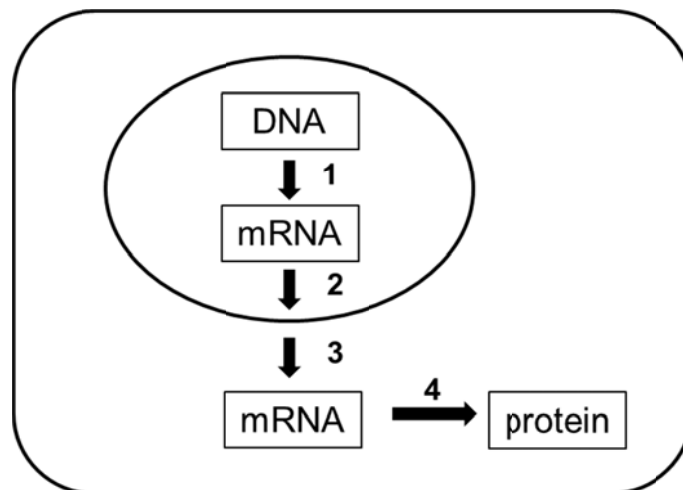
- 22 The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

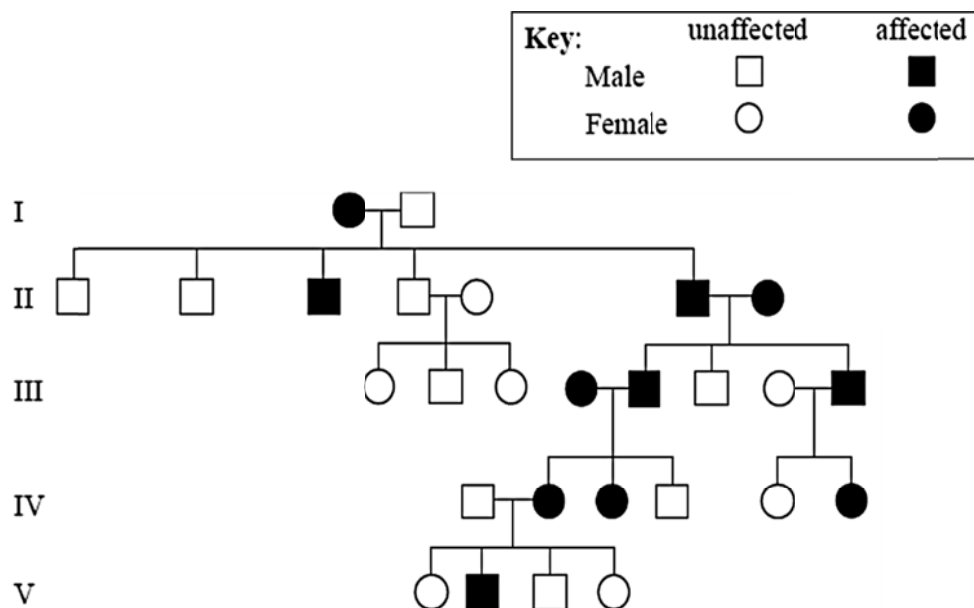
Which stage of gene expression would this mutation directly affect?

- A** Activation of the gene by a steroid hormone
B Attachment of mRNA to a ribosome
C Binding of RNA polymerase to DNA
D Splicing of pre-mRNA
- 23 The following diagram shows the expression of a particular gene to its protein product in a eukaryotic cell. Which of the following combination correctly describes steps 1 – 4?



	1	2	3	4
A	Histones are acetylated	5' capping occurs	RNase does not degrade 5' capped mRNA	Initiation factors bind to ribosome
B	Histones are methylated	Poly(A) tail is added to 3' end	Alternative splicing occurs	Initiation factors bind to ribosome
C	Activator binds to enhancer	Introns are removed	Initiation factors bind to ribosome	Ribosome binds to 5' UTR
D	Repressor does not bind to operator	Introns are removed	5' capping occurs	Initiation factors bind to ribosome

- 24 The pedigree chart below shows the inheritance of a genetic disease in a family.



What is the nature of the allele that causes this disease?

- A dominant and sex linked
- B dominant and non-sex linked
- C recessive and sex linked
- D recessive and non-sex linked

25 The statements are descriptions of aspects of genetics.

- 1 The phenotype is affected by both alleles at the same locus of a heterozygous individual.
- 2 The combined effects of alleles at two or more gene loci equal the sum of their individual effects.
- 3 Many different alleles present in a gene pool can occupy the same gene locus.
- 4 Alleles of one gene mask the effects of the alleles of another gene at a different locus.

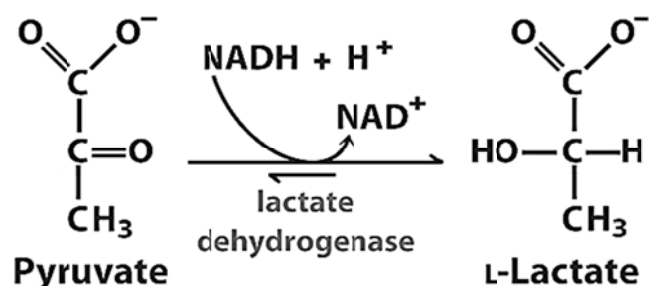
Which of the following correctly describes the statements?

	codominant alleles	epistatic alleles	multiple alleles	additive genes
A	1	4	2	3
B	1	4	3	2
C	2	1	4	3
D	3	1	4	2

26 Which one of the following is a correct outline of the main events in photosynthesis?

- A** Carbon dioxide combines with an acceptor compound and this is reduced by hydrogen split from water by light.
- B** Light joins carbon dioxide to an acceptor compound which is then reduced by hydrogen obtained from water.
- C** Light splits carbon dioxide and the resulting carbon then combines with oxygen and hydrogen obtained from water.
- D** Light splits water and the resulting hydroxyl group combines with a compound which has incorporated carbon dioxide.

- 27 The diagram shows the reversible dehydrogenase conversion of pyruvate to lactate by the enzyme lactate dehydrogenase.



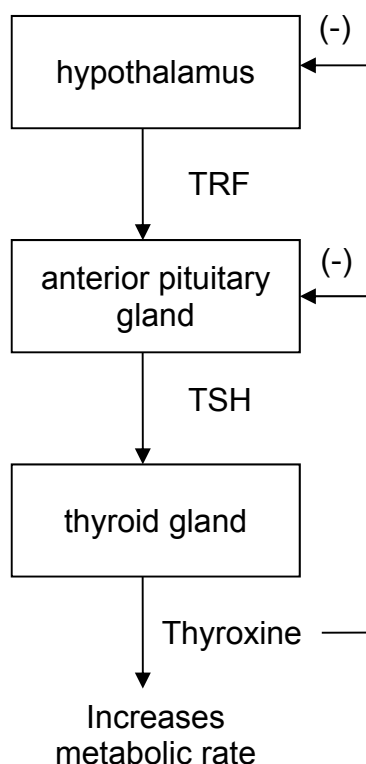
What would be the effect of inhibition of lactate dehydrogenase in a mammalian cell under anaerobic conditions?

- A Decrease in cell pH, due to the accumulation of lactic acid.
 - B Decrease in glycolysis, due to the lack of NAD.
 - C Increase in ATP production, due to increased amounts of reduced NAD.
 - D Increase in activity of the Krebs cycle, due to increased amounts of pyruvate.
- 28 From yeast cells, a mixture of enzymes can be obtained that catalyses the conversion of glucose to carbon dioxide and ethanol. Pyruvate is produced during this process.

Which combination of substances would increase the rate of ethanol production by these enzymes?

- A ATP and lactate
 - B ATP and NAD
 - C NAD and coenzyme A
 - D NAD and lactate
- 29 Which of the following statements **best** describes the outcome of the negative feedback loop in a homeostatic mechanism?
- A Blood glucose level increases to the normal level in response to the release of glucagon.
 - B Glucagon is released by the pancreas in response to a drop in blood glucose concentration below the normal level.
 - C Release of glucagon is stopped in response to blood glucose concentration returning to the normal level.
 - D The liver cells break down glycogen to glucose in response to glucagon.

- 30** Homeostasis often involves multi-step pathways. An example is given in the diagram below, which shows the hormonal feedback mechanism that controls body temperature. [TRF = thyrotrophin-releasing factor ; TSH = thyroid-stimulating hormone]

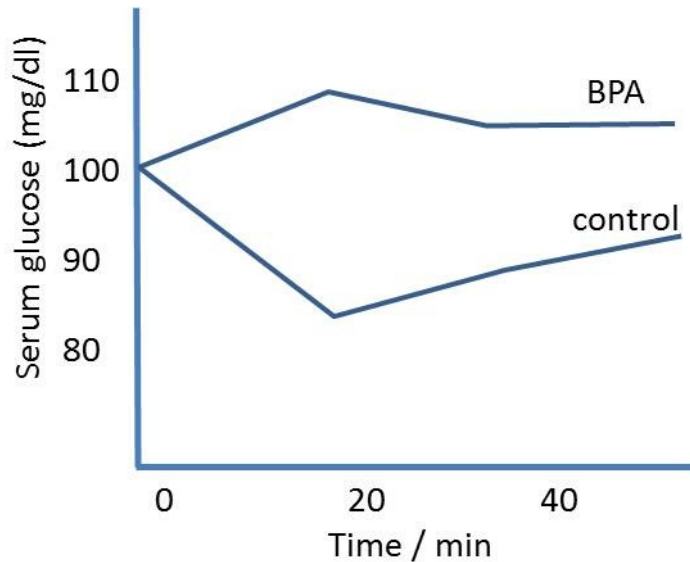


A rise in temperature causes a decrease in the amount of TRF released by the hypothalamus. As a result of this, the level of thyroxine will

- A** result in the inhibition of TRF production.
- B** result in the inhibition of TSH production.
- C** rise with a consequent fall in the basic metabolic rate.
- D** fall with a consequent fall in the basic metabolic rate.

- 31** An experiment was performed on laboratory mice to test the effects of bisphenol A (BPA), a chemical commonly used to make polycarbonate bottles. BPA was fed orally to the mice and their serum glucose levels were tested at various time intervals.

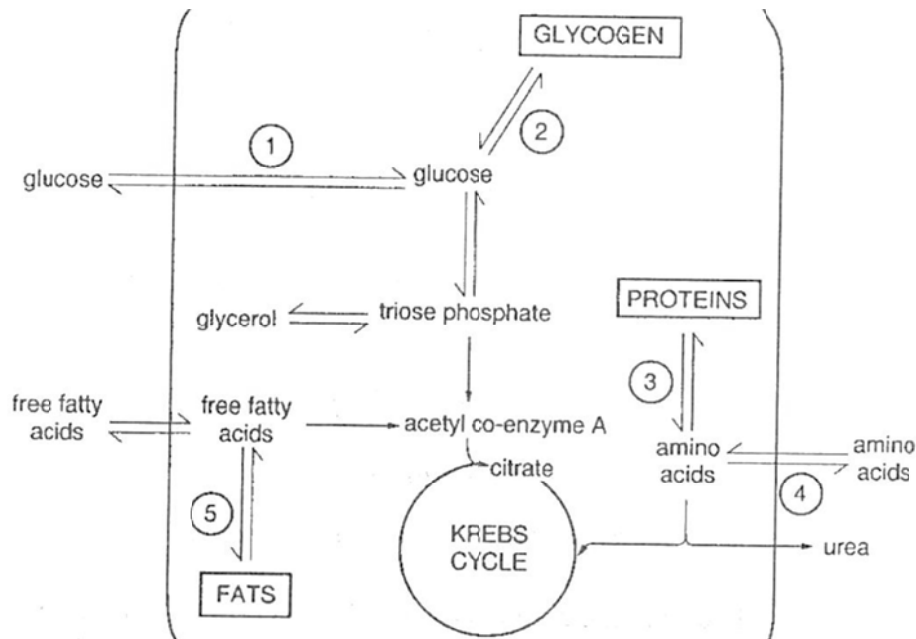
The graph shows the results of the experiment.



Based on the results shown, which of the following **cannot** be a long-term effect of BPA?

- A** BPA-treated animals will secrete more insulin than the control.
- B** Functions of β cells of the islets of Langerhans will be affected.
- C** Hypoglycaemia will develop with long term use of BPA.
- D** Insulin resistance will develop.

- 32 The diagram shows some biochemical pathways in a liver cell. Some of the points where hormones affect the pathways are labelled 1 to 5.



At which points would the hormone insulin accelerate the pathways in the directions indicated?

- A 1 and 2
 B 2 and 5
 C 1, 2 and 5
 D 2, 3 and 4
- 33 A particular type of autoimmune disease causes antibodies to destroy acetylcholine receptors of neurons. What effect will this have on the functioning of the nervous system?
- A Presynaptic neurons will be unable to release neurotransmitters into the synaptic clefts.
 B Neurons will be unable to propagate action potentials along their axons.
 C Depolarized neurons will be unable to reestablish an ionic gradient across their membranes.
 D Postsynaptic neurons will be unable to detect signals from presynaptic neurons.

- 34** Tetrodotoxin, a puffer fish toxin, blocks voltage-gated sodium channels. Black widow spider's venom causes the voltage-gated calcium channels to be constantly open. Crotoxin binds irreversibly to acetylcholine receptors.

What will happen to the nerve transmission if each toxin is applied?

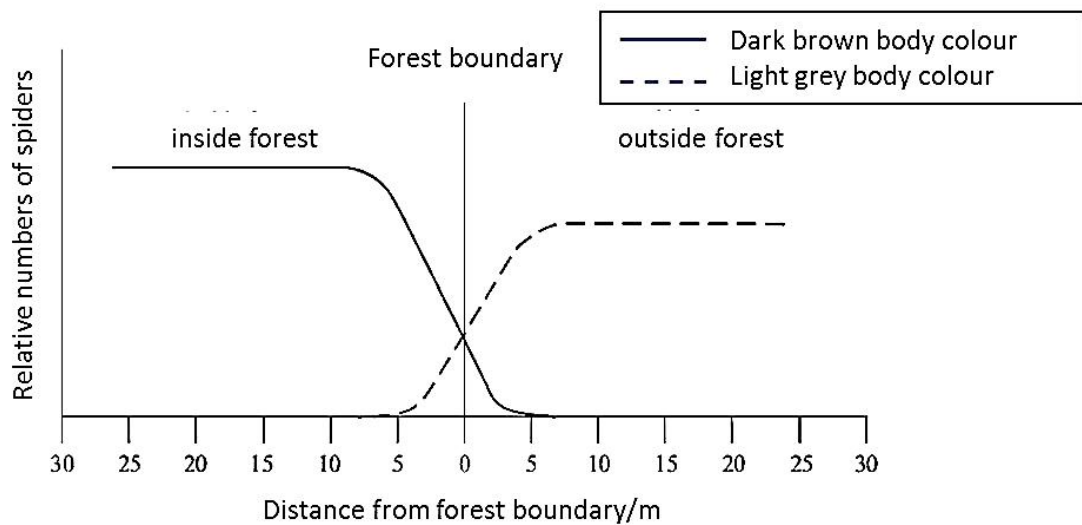
	tetrodotoxin	black widow spider's venom	crotoxin
A	block action potentials along axon	reduce transmission of impulse across synapse	increase transmission of impulse across synapse
B	block action potentials along axon	increase transmission of impulse across synapse	reduce transmission of impulse across synapse
C	increase transmission of impulse across synapse	reduce transmission of impulse across synapse	block action potentials along axon
D	reduce transmission of impulse across synapse	block action potentials along axon	increase transmission of impulse across synapse

- 35** In the summer of 1995, at least 15 iguanas survived Hurricane Marilyn on a raft of uprooted trees. They rode the high seas for a month before colonizing the Caribbean island, Anguilla. These few individuals were perhaps the first of their species, *Iguana iguana*, to reach the island.

When one such healthy iguana was mated with another from its source of origin in the Virgin Islands, the offspring were infertile. Which of the following best explains this?

- A** Allopatric speciation
- B** Artificial selection
- C** Diversifying selection
- D** Founder effect

- 36 The graph below shows the distribution of huntsman spiders at a forest boundary:



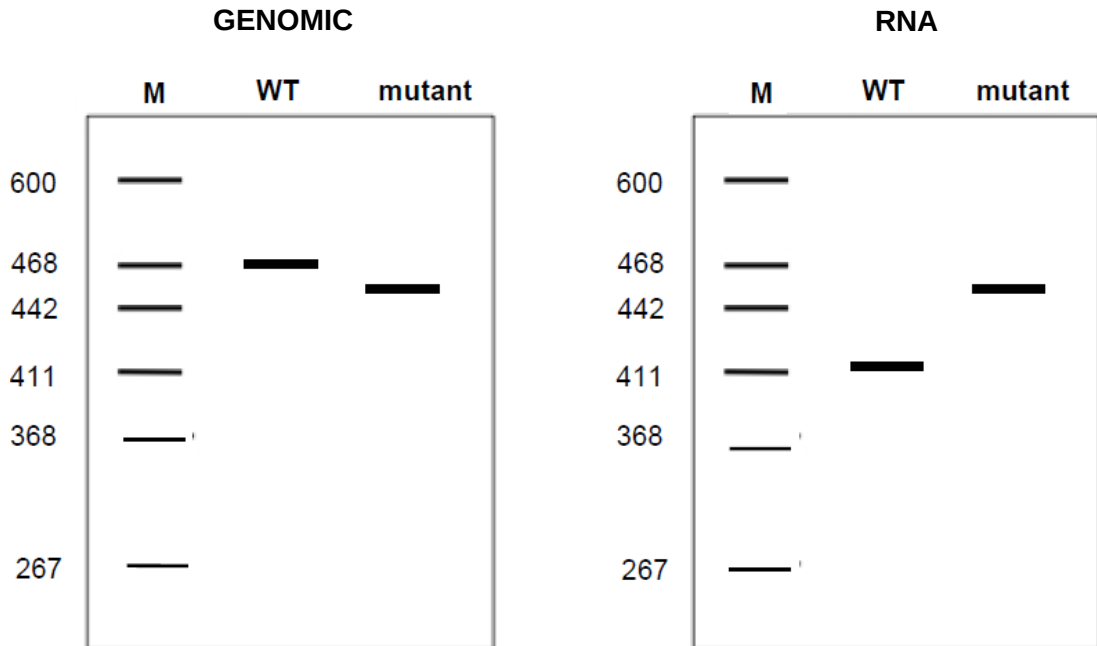
One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that the

- A differences in the two habitats had changed the physical appearance of individual spiders.
- B particular body colour provided a selective advantage to spiders in a particular habitat.
- C two populations of spiders were once different species.
- D two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.

- 37** In separate reactions, PCR was performed using a pair of primers with sequences complementary to parts of separate exons of a gene suspected to be involved in a genetic disease.

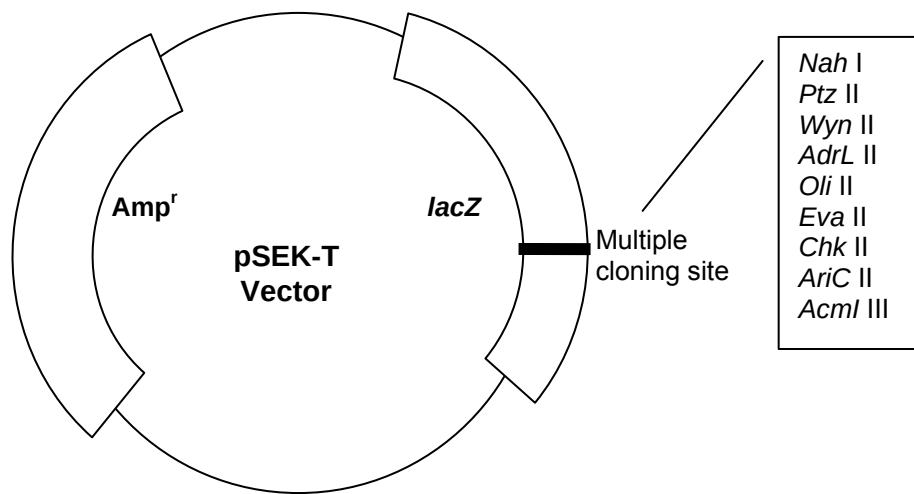
Using genomic DNA and RNA isolated from healthy (wild-type) and diseased cells, separate PCR reactions were done. The PCR products were analysed on polyacrylamide gels. The sizes of the bands in the molecular weight marker (M) are indicated in base-pairs.



Which of the following **best** explains the results obtained?

- A** Deletion of an exon
- B** Deletion of several introns
- C** Deletion of several codons
- D** Deletion of a splice site

- 38** As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the Bt toxin protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for Bt toxin were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	chemicals		colour of colonies
A	ampicillin	X-gal	blue
B	ampicillin	X-gal	white
C	β -galactosidase	X-gal	blue
D	β -galactosidase	lactose	white

- 39** One therapy to treat β -thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β -globin to the patient's cells.

Which of the following would ensure that the normal gene is passed on to the next generation?

- A** Using a retrovirus to introduce the normal β -globin gene into bone marrow cells.
 - B** Using a retrovirus to introduce the normal β -globin gene into an egg cell.
 - C** Using an adenoviral vector to introduce the normal β -globin gene into bone marrow cells.
 - D** Using an adenoviral vector to introduce the normal β -globin gene into an egg cell.
- 40** Which genetic modification could increase the yield of crop measured as mass of crop produced per unit area per year?
- A** Delayed ripening in the fruits
 - B** Herbicide resistant plants
 - C** More essential amino acids in the seeds
 - D** More vitamin A in the grain

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Qn	Answer	Qn	Answer
1	C	21	A
2	D	22	D
3	C	23	A
4	C	24	B
5	A	25	B
6	A	26	A
7	D	27	B
8	B	28	B
9	C	29	C
10	B	30	D
11	D	31	C
12	C	32	C
13	D	33	D
14	B	34	B
15	B	35	A
16	D	36	B
17	B	37	D
18	A	38	B
19	C	39	B
20	B	40	B